

Lab 9 Development Report

Application and Deployment Enhancement

CS 5542 — Big Data and Analytics
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1. PROJECT OVERVIEW

Our team has built and iteratively improved an AI-powered logistics platform across Labs 5–8. Lab 6 delivered a Multi-Agent Trucking Analytics Dashboard (Gemini 2.5 Flash + Snowflake + Streamlit). Lab 7 reproduced and extended the ReMindRAG knowledge-graph RAG system. Lab 8 introduced HyperLogistics — a domain-adapted safety validation agent (Phi-2 + QLoRA) for explainable rerouting under real-time disruptions. Lab 9 directly addresses gaps and instructor feedback from all three prior labs and advances both systems toward Research-A-Thon demonstration readiness.

GitHub (Labs 5–6, 8–9): https://github.com/mosomo82/COMP_SCI_5542

GitHub (Lab 7 — ReMindRAG): https://github.com/mosomo82/ReMindRAG_Week7

Lab 6 Live App: <https://cs5542logisticsai.streamlit.app/>

Lab 8 Live App: <https://cs5542lab8.streamlit.app/>

2. PRIOR LAB FEEDBACK ADDRESSED

Lab 9 responds to instructor feedback received on Labs 6 and 7, and proactively addresses the critical evaluation and documentation gaps identified in the Lab 8 submission.

Lab 6 Feedback

■ *"Minor improvements could include expanding the documentation of system architecture and evaluation details."*

Response: ARCHITECTURE.md and EVALUATION.md authored and committed to Week_9/. See Section 3.5.

Lab 7 Feedback

■ *"Minor improvements could include expanding the benchmarking results and adding additional automated verification for reproducibility experiments."*

Response: BENCHMARKING.md (constrained LooGLE + HotpotQA runs) and REPRO_VERIFICATION.md (suite expanded 17 → 26 tests with CI) authored and committed to ReMindRAG_Week7/. See Section 3.6.

Lab 8 — Proactive Improvements

Lab 8 identified three critical issues in its own evaluation: a monotonicity test failure (APPROVE not flipping to VETO on bridge violation), hallucinated bridge constraints on local hardware, and a contradictory accuracy number in the report. Lab 9 addresses all three, plus expands the evaluation set, replaces binary scoring with a rubric, and adds CI integration. See Section 3.7.

3. APPLICATION ENHANCEMENTS (LAB 9)

3.1 Improved Streamlit User Interface (Lab 6)

- Persistent sidebar with data-freshness indicator and quick-link navigation across all nine dashboard tabs.
- Agent Chat tab redesigned with message history, loading spinners, and collapsible tool-call trace expander (tool name, arguments, latency per step).
- Input validation and user-friendly error messages in the Executive tab SQL Explorer.
- Reset Session button to clear Streamlit state and Gemini conversation history without a page reload.

3.2 Monitoring and Evaluation Features (Lab 6)

- Agent evaluation suite expanded from 5 to 10 scenarios with CI integration and a 70% pass-rate quality gate.
- Real-time latency percentile cards (p50 / p90 / p99) added to the Monitoring tab.
- Lab 7 ReMindRAG smoke test integrated into the same GitHub Actions CI workflow.

3.3 Logging and Debugging Support (Lab 6)

- Python logging module replaces ad-hoc print statements (INFO default; DEBUG via LOG_LEVEL env var).
- Startup assertions in sf_connect.py validate Snowflake parameters at import time.
- Tool-call trace expander in Agent Chat tab surfaces each agent step with tool name, arguments, and latency.

3.4 Deployment and System Stability (Lab 6)

- Snowflake warehouse keep-alive ping reduces median cold-start latency from ~8 s to under 2 s.
- Exponential-backoff retry decorator on all Gemini API calls (5 s → 10 s → 20 s, max 3 retries).
- FastAPI /health endpoint + UptimeRobot integration for continuous uptime monitoring.
- Full dependency pinning (146 packages) validated against clean Docker build.

3.5 Lab 6 Documentation Expansion

- ARCHITECTURE.md — Four-layer system architecture, agent lifecycle, nine analytical tools, observability stack, env variables, limitations/future-work.
- EVALUATION.md — Baseline results, S2 root-cause analysis, 10-scenario expanded suite, CI methodology, eval_results.json schema, traceability table.

3.6 Lab 7 Benchmarking and Automated Verification

- BENCHMARKING.md — Paper-vs-reproduction comparison; constrained 5-title LooGLE run (F1 = 0.496, consistent with paper's 0.49); HotpotQA subset run (F1 = 0.58 vs paper's 0.61); efficiency verification; variance table; JSON output schema.
- REPRO_VERIFICATION.md — Nine new automated tests in two new files (test_benchmark_smoke.py: 5 tests; test_repro_variance.py: 4 tests); suite expanded 17 → 26 tests; full CI integration at zero API cost.

3.7 Lab 8 Critical Fixes and Evaluation Expansion

- CPP hard gate (Spatial SQL): Bridge constraint check is now deterministic SQL before LLM invocation. Monotonicity invariant (APPROVE → VETO on bridge violation) is enforced by the query, not the model — resolving the Q13→Q14 failure on both Colab and RTX 3060.
- Evidence injection: Bridge constraints are retrieved from SILVER.BRIDGE_INVENTORY_GEO and injected verbatim into the LLM prompt as a [RETRIEVED CONSTRAINTS] block. Model is instructed not to recall physical limits from memory — hallucination rate drops from ~40% to 8%.
- Accuracy discrepancy corrected: Section 5 intro contained a contradictory baseline accuracy number (60% vs 40%). EVALUATION.md documents the correct values (40% baseline on both platforms) and

explains the Colab vs RTX 3060 PEFT discrepancy.

- CoT-formatted training data: The dataset generation script is updated to produce full 4-step reasoning chains (Disruption → Route → Constraint → Decision) in all output fields. Retraining with 300+ pairs (expanded from 100) narrows the PEFT CoT quality gap from 67% to 83%.
- Evaluation set expanded 15 → 50 queries across 5 disruption categories. Binary APPROVE/VETO scorer replaced with a 5-dimension rubric (Decision Correctness, Disruption Grounding, Constraint Citation, CoT Completeness, Jargon Accuracy) scored 0–10.
- Metamorphic test suite expanded from 3 to 13 pairs, including two additional monotonicity tests for weight limits and height clearances, and a regression test for the original Q13→Q14 failure.
- GitHub Actions CI added: evaluation harness runs in mock mode (no GPU) on every push to main, with a 70% pass-rate quality gate consistent with Labs 6 and 7.
- ARCHITECTURE.md and EVALUATION.md authored for Week_8/, documenting the 4-layer CPP architecture, failure mode analysis, and fix methodology.

4. EXTENSION OF THE PHASE-2 PROTOTYPE

Lab	Capability	Phase-2 Baseline	Lab 9 Enhancement
6	Architecture docs	README only	ARCHITECTURE.md + EVALUATION.md
6	UI & monitoring	Nine raw tabs, manual eval (5 s/cr)	Side bar, tool trace, 10 scen., CI
6	Logging & stability	Print stmts, no retry, 8s cold-start	Structured logging, retry, <2s
7	Benchmarking	Metrics unverified (quota limited)	5-title LooGLE + HotpotQA runs
7	Repro. verification	17 structural tests (mock agents)	26 tests incl. output + determinism
8	Monotonicity failure	Q13→Q14 FAIL on both platforms	PASS — CPP Spatial SQL hard gate
8	Hallucination (PEFT)	~40% rate on local hardware	8% — evidence injection in prompt
8	Accuracy discrepancy	60% vs 40% contradiction in reports	Corrected — 40% both platforms
8	CoT quality (PEFT)	67% (training data not CoT-formatted)	83% — 300+ CoT-formatted pairs
8	Evaluation coverage	15 queries, binary scorer	50 queries, 5-dim rubric (0–10)
8	CI integration	None	GitHub Actions mock-mode eval
8	Architecture docs	Group report only	ARCHITECTURE.md + EVALUATION.md

5. REPOSITORY AND DEPLOYMENT LINKS

Resource	URL
Lab 6 Live App	https://cs5542logisticsai.streamlit.app/
Lab 8 Live App	https://cs5542lab8.streamlit.app/
Main Repo (Labs 5–6, 8–9)	https://github.com/mosomo82/COMP_SCI_5542
ReMindRAG Repo (Lab 7)	https://github.com/mosomo82/ReMindRAG_Week7
Lab 6 ARCHITECTURE.md	github.com/.../Week_9/ARCHITECTURE.md

Lab 6 EVALUATION.md	github.com/.../Week_9/EVALUATION.md
Lab 7 BENCHMARKING.md	github.com/mosomo82/ReMindRAG_Week7/blob/main/BENCHMARKING.md
Lab 7 REPRO_VERIFICATION.md	github.com/mosomo82/ReMindRAG_Week7/blob/main/REPRO_VERIFICATION.md
Lab 8 ARCHITECTURE.md	github.com/.../Week_8/ARCHITECTURE.md
Lab 8 EVALUATION.md	github.com/.../Week_8/EVALUATION.md

6. SUMMARY

Lab 9 closes gaps across all three prior labs. Lab 6 feedback is addressed by ARCHITECTURE.md and EVALUATION.md. Lab 7 feedback is addressed by BENCHMARKING.md and REPRO_VERIFICATION.md (17 → 26 automated tests). Lab 8's critical failures — monotonicity, hallucination, and the accuracy discrepancy — are resolved by the CPP Spatial SQL hard gate, evidence injection, and corrected reporting. The evaluation suite expands from 15 to 50 queries with rubric scoring and CI integration. All improvements are documented in per-lab ARCHITECTURE.md and EVALUATION.md artifacts, positioning both systems for the Research-A-Thon competition.